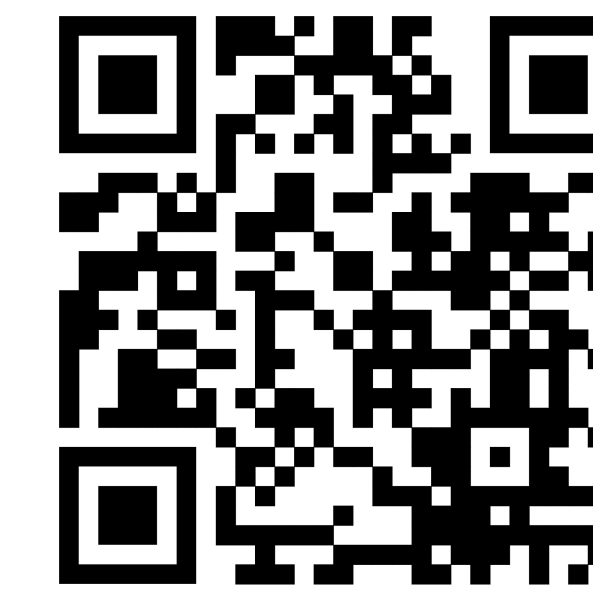




Variational Inference and Natural Gradient VI

Problem Setting:

$$q^*(z) = \operatorname{argmin}_{q \in \mathcal{Q}} D_{\text{KL}}(q(z) \parallel p(z \mid \mathcal{D})).$$

Non-conjugate setting: $p(z \mid \mathcal{D}) \propto p(\mathcal{D} \mid z)p(z)$ is intractable.

Equivalent to minimizing negative Evidence Lower BOund (ELBO):

$$\ell := -\mathbb{E}_{q(z)}[\log p(\mathcal{D} \mid z)] + D_{\text{KL}}(q(z) \parallel p(z)).$$

Suppose $\mathcal{Q}$ is parameterized by $\theta \in \Theta$.

VI update with SGD:

$$\theta_{t+1} = \operatorname{argmin}_{\theta \in \Theta} \left\langle \tilde{\nabla} \ell(\theta_t), \theta - \theta_t \right\rangle + \frac{1}{2\gamma_t} \|\theta - \theta_t\|^2.$$

Natural Gradient VI (NGVI) update with Stochastic Natural GD (SNGD):

$$\theta_{t+1} = \operatorname{argmin}_{\theta \in \Theta} \left\langle \tilde{\nabla} \ell(\theta_t), \theta - \theta_t \right\rangle + \frac{1}{\gamma_t} D_{\text{KL}}(q(z; \theta) \parallel q(z; \theta_t)).$$

Advantages of NGVI:

- ✓ Suitable *KL geometry* instead of Euclidean.
- ✓ *Interpretable* as Mirror Descent (in expectation parameters).
- ✓ Leads to *considerable improvements* in practice [1].

Challenge: For non-conjugate models, both *convexity* and *smoothness* fail, e.g., logistic or Poisson regression [1]. Theoretical guarantees for NGVI in non-conjugate setting remain an *open problem*.

NGVI and Mirror Descent

Let $\mathcal{Q}$ be an exponential family indexed by η :

$$q(z; \eta) = h(z) \exp(\langle \phi(z), \eta \rangle - A(\eta)).$$

For exponential family $\mathcal{Q}$, NGVI update is equivalent to

$$\omega_{t+1} = \operatorname{argmin}_{\omega \in \Omega} \langle \nabla \hat{\ell}(\omega_t), \omega - \omega_t \rangle + \frac{1}{\gamma_t} D_{A^*}(\omega, \omega_t).$$

Here A^* is the convex conjugate of A , and D_{A^*} is the Bregman divergence.

NGVI for Mean-Field Gaussian Family:

Natural parameter: $\eta = (\lambda, \Lambda) = (\Sigma^{-1}\mu, -\frac{1}{2}\Sigma^{-1})$.

Expectation parameter: $\omega = (\xi, \Xi) = (\mu, \Sigma + \mu\mu^\top)$.

Log-partition function: $A(\eta) = -\frac{1}{4}\lambda^\top \Lambda^{-1}\lambda - \frac{1}{2}\log \det(-2\Lambda)$.

References

- [1] Kaiwen Wu and Jacob R. Gardner. Understanding Stochastic Natural Gradient Variational Inference. ICML, 2024.
- [2] Haihao Lu, Robert M Freund, and Yurii Nesterov. Relatively smooth convex optimization by first-order methods, and applications. SIAM Journal on Optimization, 2018.
- [3] Ilyas Fatkhullin, Niao He, and Yifan Hu. Stochastic optimization under hidden convexity. SIAM Journal on Optimization, 2025.
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Contribution I: Landscape Properties

Relative smoothness [2]:

A differentiable function ℓ is said to be L -smooth relative to A^* for some $L > 0$ if for any $u, v \in U$,

$$-LD_{A^*}(v, u) \leq \ell(v) - \ell(u) - \langle \nabla \ell(u), v - u \rangle \leq LD_{A^*}(v, u).$$

This is a *generalization of smoothness in non-Euclidean geometry*.

ELBO is A^* -Relatively Smooth

Theorem 1. Let $f(z) := -\log p(\mathcal{D} \mid z) \in \mathcal{C}^4$ on $\mathbb{R}^d$, and $|\nabla_i f(z)| \leq L_1$, $|\nabla_{ij}^2 f(z)| \leq L_2$. Then $\ell(\omega)$ is L -smooth relative to A^* on $\tilde{\Omega} := \{\omega : |\mu_i| \leq U, D^{-1} \leq \Sigma_{ii} \leq D\}$ with $L = \mathcal{O}(dD^2U(L_1 + L_2U) + dD^3(L_1 + L_2U))$.

Hidden convexity [3]:

A function F is hidden convex with modulus $\mu_C > 0, \mu_H \geq 0$ if it has a *convex reformulation* $F(x) = H(c(x))$, where

1. the map c is invertible satisfying

$$\|c(x_1) - c(x_2)\| \geq \mu_C \|x_1 - x_2\|.$$

2. H is μ_H -strongly convex.

ELBO is Hidden Convex

Theorem 2. If $-\log p(\mathcal{D} \mid z)$ is convex in z , then $\ell(\omega)$ is hidden convex on $\tilde{\Omega}$ with $\mu_C = (4U^2 + 4D + 1)^{-1/2}$ and $\mu_H = 1$. Important implication: Every stationary point is a *global minimum*!

Contribution II: Non-Euclidean Projection for NGVI

We have proved relative smoothness and hidden convexity of the objective on the bounded domain $\tilde{\Omega} = \{\omega : |\mu_i| \leq U, D^{-1} \leq \Sigma_{ii} \leq D\}$.

Proj-SNGD Algorithm (NGVI with Projection):

$$\nabla A^*(\omega_{t+1,*}) = \nabla A^*(\omega_t) - \gamma_t \tilde{\nabla} \ell(\omega_t), \quad \omega_{t+1} = \operatorname{Proj}_{\tilde{\Omega}}(\omega_{t+1,*}).$$

Implementation of $\operatorname{Proj}_{\tilde{\Omega}}(\omega_{t+1,*})$:

1. Compute $(\mu_{t+1,*}, \Sigma_{t+1,*})$ which corresponds to $\omega_{t+1,*}$.
2. Compute $(\mu_{t+1}, \Sigma_{t+1})$ by entry-wise clipping of $(\mu_{t+1,*}, \Sigma_{t+1,*})$.
3. Compute the corresponding expectation parameter ω_{t+1} of $(\mu_{t+1}, \Sigma_{t+1})$.

Only $\mathcal{O}(d)$ extra arithmetic operations for projection.

Contribution III: Convergence of Proj-SNGD

Stationary Criterion: Bregman Forward-Backward Envelope (BFBE) [4]: $\mathcal{E}_\rho(\omega) := -2\rho \min_{\omega' \in \tilde{\Omega}} [\langle \nabla \ell(\omega), \omega' - \omega \rangle + \rho D_{A^*}(\omega', \omega)]$ for some $\rho > 0$.

Stationary Point Convergence

Theorem 3. Under relative smoothness condition and other mild conditions on the likelihood, the iterates of Proj-SNGD satisfy

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\mathcal{E}_{3L}(\omega_t)] \leq 18 \frac{L\lambda_0}{T} + 9 \sqrt{\frac{LV^2\lambda_0}{T}},$$

where $\lambda_0 := \ell(\omega_0) - \ell^*$, and V^2 is the norm-invariant variance of $\tilde{\nabla} \ell(\omega)$.

$\mathcal{O}(1/\sqrt{T})$ in the noisy setting. $\mathcal{O}(1/T)$ in the deterministic setting.

Global Convergence

Theorem 4. Assume in addition that $-\log p(\mathcal{D} \mid z)$ is convex, then under some mild assumptions the iterates of Proj-SNGD satisfy

$$\min_{t \leq T-1} \mathbb{E}[\ell(\omega_{t,*}) - \ell^*] \leq \frac{192L\lambda_0}{\mu_B} \exp\left(-\frac{\mu_B T}{12L}\right) + \frac{648LV^2}{\mu_B^2 T},$$

where $\omega_{t,*} := \operatorname{argmin}_{\omega' \in \tilde{\Omega}} \langle \nabla \ell(\omega_t), \omega' \rangle + LD_{A^*}(\omega', \omega_t)$, $\mu_B := \frac{\mu_C^2}{9U^2D^2}$.

$\mathcal{O}(1/T)$ in the noisy setting. **Linear rate** in the deterministic setting.

Experiment Results

Problem: Bayesian logistic regression on MNIST dataset.

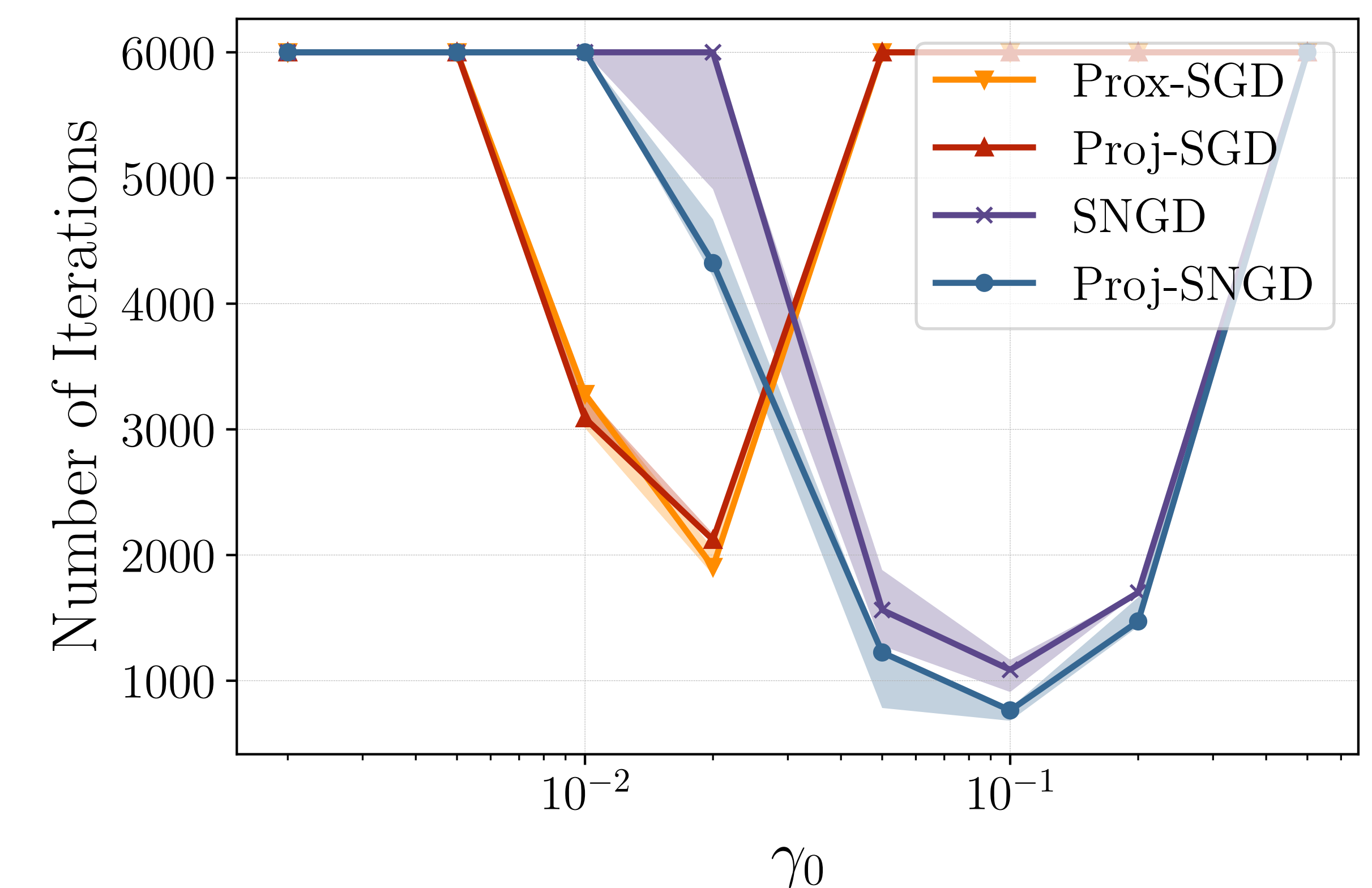


Figure 1: Number of iterations before the objective falls below $\ell(\omega) \leq 700$ for different initial step sizes γ_0 . Prox-SGD and Proj-SGD: standard VI algorithms. SNGD: NGVI without projection. Proj-SNGD: this paper.

- ✓ NGVI algorithms converge faster and are more robust to step size tuning.
- ✓ Projection accelerates convergence.